2.7.28: Area of a Triangle

Taraka Abhinav - EE25BTECH11016

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Question

Find the area of the triangle whose vertices are $A(2,3)$, $B(3,5)$, and $C(4,4)$.

Method: Area using Determinant

The area of a triangle with vertices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ can be found using the formula:

$$\text{Area} = \frac{1}{2}\left|\det(\mathbf{A}-\mathbf{B}, \mathbf{A}-\mathbf{C})\right|$$

The given vertices are:

$$\mathbf{A} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

Step 1: Find the Side Vectors

First, we calculate two vectors that form the sides of the triangle.

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix} \tag{1}$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \tag{2}$$

Step 2: Calculate the Determinant

Next, we find the determinant of the matrix formed by these two vectors.

$$\begin{aligned} \det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C}) &= \begin{vmatrix} -1 & -2 \\ -2 & -1 \end{vmatrix} \\ &= (-1)(-1) - (-2)(-2) \\ &= 1 - 4 = -3 \end{aligned} \tag{3}$$

Step 3: Find the Area

The area is half the absolute value of the determinant.

$$\text{Area} = \frac{1}{2}|-3| = 1.5 \tag{4}$$

Thus, the area of the triangle is 1.5 square units.

C Code - Verification (Cross Product)

```c
#include <stdio.h>
#include <math.h>

// Function to compute area via cross product
double area_triangle(double A[3], double B[3], double C[3]) {
    double AB[3] = {B[0] - A[0], B[1] - A[1], 0};
    double AC[3] = {C[0] - A[0], C[1] - A[1], 0};
    double cross_z = AB[0]*AC[1] - AB[1]*AC[0];
    return 0.5 * fabs(cross_z);
}

int main() {
    double A[3] = {2,3,0};
    double B[3] = {3,5,0};
    double C[3] = {4,4,0};

    double area = area_triangle(A,B,C);
    printf("Area of triangle = %.2f\n", area);
    return 0;
```

Python + C Code

```
import ctypes
import numpy as np

# Load compiled C library
# gcc -shared -o triangle_area.so -fPIC area.c
lib = ctypes.CDLL("./triangle_area.so")

# Define argument and return types
lib.area_triangle.argtypes = [ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.c_double)]
lib.area_triangle.restype = ctypes.c_double

# Define points as 3D vectors
A = (ctypes.c_double * 3)(2,3,0)
B = (ctypes.c_double * 3)(3,5,0)
C = (ctypes.c_double * 3)(4,4,0)

# Call C function
```

Pure Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Define points as 3D vectors
A = np.array([2,3,0])
B = np.array([3,5,0])
C = np.array([4,4,0])

# Form side vectors
vec_AB = B - A
vec_AC = C - A

# Cross product & area
cross_product = np.cross(vec_AB, vec_AC)
area = 0.5 * np.linalg.norm(cross_product)
print(f"Area of triangle = {area:.2f}")
```

Python Plotting Code

```python
# Plot the triangle
# Slicing [:2] to plot the 2D coordinates
triangle = np.array([A[:2], B[:2], C[:2], A[:2]])
plt.plot(triangle[:, 0], triangle[:, 1], 'bo-')

# Add labels
plt.text(A[0]-0.3, A[1], f'A({A[0]},{A[1]})')
plt.text(B[0]+0.1, B[1], f'B({B[0]},{B[1]})')
plt.text(C[0]+0.1, C[1], f'C({C[0]},{C[1]})')
plt.text(3, 3.5, f'Area = {area:.2f}', color='red')

plt.xlabel("X-axis"); plt.ylabel("Y-axis")
plt.title("Triangle ABC"); plt.grid(True)
plt.gca().set_aspect('equal', adjustable='box')
plt.savefig("figs/Figure_1.png"); plt.show()
```

Plot of the Triangle